EXPIREMENT-02

Ex.No.2	**Study of Network cables.**
Date:	1. Understand different types of Network cables. **2.** Make a cross-wired cable and straight through cable using clamping/crimping tool.

a. <u>Understand different types of network cable.</u>

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

Cable type	**Category**	**Maximum Data Transmission**	**Advantages/ Disadvantages**	**Application/Use**	**Image**
UTP	Category 3	10 bps	**<u>Advantages</u>** • Cheaper in cost • Easy to install as they have a smaller overall diameter. **<u>Disadvantages</u>** • More prone to (EMI) Electromagnetic interference and noise	10Base-T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet	
	Category 5	Up to 100 Mbps			
	Category 5e	1Gbps			
STP	Category6,6a	10Gbps	**<u>Advantages</u>** • Shielded. • Faster than UTP. • Less susceptible to noise and interference	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres	

SSTP	Category 7	10Gbps	**Disadvantages** • Expensive • Greater installatio n effort	Gigabit Ethernet, 10G Ethernet (100m)	

Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile • **Disadvantages** • Limited distance • Cost • Size is bulky	Speed of signal is 500m Television network High speed internet connections
fibre optics cable	Single mode Multi mode	100Gbps	**Advantages** • High speed • High bandwidth • High security • Long distance **Disadvantages** • Expensive • Requires skilled installers	• Maximum distance of fibre optics cable is around 100meters

b. <u>Make Your Own Ethernet Cross-Over Cable/ Straight cable</u>

Tools and parts needed:

- Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs.
- Optional two plug shields.

- **Ethernet Cable (CAT5e/CAT5)**

- CAT5e is recommended because it supports Gigabit Ethernet (1000 Mbps).
- CAT5 also works but supports lower speeds and shorter distances.

- **Crimping Tool**

  - A networking tool used to:
    - Strip the outer insulation of the cable.
    - Arrange and trim the wires.
    - Crimp (press) the RJ45 connector onto the cable.

- **RJ45 Connectors (2)**

  - Plastic connectors fitted at both ends of the Ethernet cable.
  - Used to connect the cable to computers, switches, routers, etc.

- **Plug Shields (Optional)**

  - Protective covers placed over the RJ45 connectors.
  - Prevent damage to the locking clip and reduce strain on the cable.

RESULT:

How to Make an Ethernet Cable (Straight-Through / Cross-Over)

Tools & Parts


CAT5e / CAT6 Cable


RJ45 Connector


Crimping Tool


Cable Stripper (Optional)

1 Cut the Cable

Cut the Ethernet cable to the desired length.



2 Strip the Outer Jacket

Use the cable stripper to remove about 2–3 cm (1 inch) of the outer jacket.



3 Untwist and Straighten the Wires

Untwist the wires and straighten them.



4 Arrange the Wires (Choose a Standard)

Straight-Through Cable: Use the same standard on both ends (T568B or T568A).



5 Trim the Wires

Trim the wires evenly so that they are all the same length.



6 Insert into RJ45 Connector

Insert the wires into the RJ45 connector. Make sure each wire goes all the way to the end and the order is correct.



7 Crimp the Connector

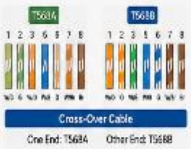
Place the connector into the crimping tool and squeeze firmly to crimp it onto the cable.



8 Repeat on the Other End

Repeat the same steps on the other end of the cable.

- For Straight-Through: Use the same standard as the first end.
- For Cross-Over: Use T568A on one end and T568B on the other.



9 Test the Cable

Use a cable tester to check if all 8 wires are connected properly.



10 Finished

Your Ethernet cable is ready to use!



Straight-Through Cable (Same Standard on Both Ends)



Cross-Over Cable (T568A to T568B)



Tips

- Make sure the wire order is correct.
- Push wires all the way to the end of the RJ45 connector.
- Crimp firmly for a good connection.